

- 4.7.11.** Let T be the statement: The sum of any two rational numbers is rational. Then T is true, but the following “proof” is incorrect. Find the mistake.

Proof by contradiction: Suppose not. That is, suppose that the sum of any two rational numbers is not rational. This means that no matter what two rational numbers are chosen their sum is not rational. Now both 1 and 3 are rational because $1 = 1/1$ and $3 = 3/1$, and so both are ratios of integers with a nonzero denominator. Hence, by a supposition, the sum of 1 and 3, which is 4, is not rational. But 4 is rational because $4 = 4/1$, which is a ratio of integers with a nonzero denominator. Hence 4 is both rational and not rational, which is a contradiction. This contradiction shows that the supposition is false, and hence statement T is true.

Suppose the sum of any two rational numbers is rational to be true. Let T be the given statement.

We must find the mistake in the contradiction.

Let's find the argument form of T :

$$T = \forall r, s \in \mathbb{Q}, \quad r + s \in \mathbb{Q}.$$

If we negate statement T :

$$\neg T = \exists r, s \in \mathbb{Q} \text{ such that } r + s \notin \mathbb{Q}.$$

But the proof states:

“Suppose not. That is, suppose that the sum of any two rational numbers is not rational,” which can be written as:

$$\neg T = \forall r, s \in \mathbb{Q}, \quad r + s \notin \mathbb{Q}.$$

Which is an incorrect negation.

Solution: The proof does not fulfill the definition of a proof by contradiction, since the statement is not properly negated.